

File Paths in R

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File Paths

This is a tutorial on file paths and directories (folders).

A directory indicates the location on the computer that you are saving your information, usually on your hard drive. A path is a slash-separated list of directory names, followed by a file name or a directory (folder) name. For example, `/Users/robertgulotty` suggests that there is a folder named `robertgulotty` in the `Users` folder.

If you look at the *file path* of some file in your computer, by right clicking on the file and selecting properties or get info, you will find that directory names are separated by `/` in Macs and other Unix based systems and `\` in Windows (PC).

These slashes differ across Macs and PCs, and R has trouble with backslashes (`\`). If you use R on a mac or a PC, you will see the path printed with a forwardslash (`/`). This is because R is built for Unix, and backslash is a programming command in the R language.

On Windows machines to tell your computer where to look, you must either

- use the forwardslash (`/`)
- use *two* backslashes (`\\`).

Either one will work.

Working Directory

The following command tells us what the ‘working directory’ is:

```
getwd()
```

```
## [1] "/Users/gulotty/Dropbox/mathCamp/Code/Rmarkdown"
```

A working directory is the default place for R to look for files and folders. If you read out data into a csv, you will create a file in the working directory.

If my working directory is “`/Users/robertgulotty/Google Drive/stuffforclass/`” the following three commands are the same:

```
# WBdta<-read.csv("/Users/robertgulotty/Google Drive/stuffforclass/WB_dta.csv")
# WBdta<-read.csv("WB_dta.csv")
# WBdta<-read.csv("~/Google Drive/stuffforclass/WB_dta.csv")
```

In the Mac operating systems, a directory usually begins with `/Users/yourname` This can be replaced with a `~`, which is a shortcut for `/Users/yourname`

In Windows operating systems, a directory usually begins with `C:/Users/yourname/`

setwd() to change the Working Directory

You can change directories as follows:

```
## Mac change directory
# setwd("/Users/robertgulotty/Google Drive/IPS2014/")
## PC change directory (Version 1)
# setwd("C:/Users/Google Drive/IPS2014/")
## PC change directory (Version 2)
# setwd("C:\\Users\\Google Drive\\IPS2014\\")
```

Where the bit in quotation marks is replaced by your preferred file. Note that you can use the tab key to complete the file name!

Expert Tip

If you need to share code, and you are often switching between different operating systems, remember to set your working directory at the top of the R script. This will avoid having to write the path later on.
